MILITARY SPECIFICATION

TRACER, PROJECTILE, M3A1, ASSEMBLY, PARTS FOR, AND LOADING, ASSEMBLING AND PACKAGING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the parts for, and the loading, assembling and packaging for one type of tracer designated as Tracer, Projectile, M3A1, Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-601	-Boxes, Wood, Cleated-Plywood
PPP-B-621	-Boxes, Wood, Nailed and Lock-corner
PPP-B-636	-Boxes, Shipping, Fiberboard
PPP-B-676	-Boxes, Setup

Military

MIL-A-2550	-Ammunition, General Specification For
MIL-B-117	-Bags, Sleeves and Tubing, - Interior Packaging
MIL-P-223	-Powder, Black
MIL-P-116	-Preservation-Packaging-Methods of

STANDARDS

Military

MIL-STD-105	-Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	-Quality Assurance Terms and Definitions
MIL-STD-414	-Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-1167	-Ammunition Data Card
MIL-STD-1168	-Lot Numbering of Ammunition

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2.1 (Cont'd)

MIL-STD-1234	-Pyrotechnics: Sampling, Inspection and Testing
MIL-STD-1235	-Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

Armament Command	
D11751175	-Tracer, Projectile, M3A1, Assembly
C8796522	-Marking Diagram and Sealing for Wood Packing Boxes
LI-11751175	-Index of Inspection Equipment Lists for Tracer, Projectile, M3A1, Assembly

2.2 Other Publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ANSI-B46.1	-Surface Texture

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, N.Y. 10018).

ASTM Methods	-American Society for Testing and Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

3. REQUIREMENTS

3.1 General.

3.1.1 Tracer and igniter compositions.-The materials used in the tracer and igniter compositions shall comply with the requirements specified on the drawings and referenced specifications (See 3.7.1, 3.7.2, 4.3.3.1, 4.4.1 & 6.4).

3.1.2 Initiator charge and relay ignition charge.-The black powder used in these charges shall comply with MIL-P-223 (see 3.7.3, 4.3.3.2, 4.4.2 and 6.5).

3.1.3 Metal Parts.-The materials, parts and assemblies shall comply with the requirements specified on the applicable drawings and referenced specifications.

3.2 Heat treatment.-Cup, Initiator.-The initiator cup shall be stress relief annealed after forming.

3.3 Residual stresses.-Cup, Initiator.-The initiator cup shall be free of residual stresses to the degree detectable by a mercurous nitrate test (see 4.3.3.5 & 4.4.5).

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3.4 Metal defects.

3.4.1 Body.-The body, including the cavity, shall be free from cracks, splits, bursts, cold shuts, pipe, porosity, inclusions, folds, seams, and other metal defects.

3.4.2 Body cavity.-The body cavity shall be free of scale, fins, burrs, draw marks, laminations, imbedded foreign matter, pits, and sharp edges.

3.4.3 Visual standards.-Visual standards for metal defects will be established for each contract after the first months production. However, the establishment of visual standards will not waive the requirement for compliance with 3.4.1 and 3.4.2 (see 6.2).

3.5 Workmanship.-The requirements for workmanship are as shown on the applicable drawings, in the applicable specifications and the following:

3.5.1 Surface finish -The requirements for surface finish are as detailed on the applicable drawing.

3.5.2 Threads.-Threads shall be full and undamaged for the entire minimum length as required by the applicable drawings and shall be in accordance with the provisions of MIL-A-2550.

3.5.3 Burr.-No part shall have a burr which might interfere with the assembly or function of the round or which might be injurious to personnel handling the item.

3.5.4 Foreign matter.-No part or assembly shall contain dirt, grease, chips, rust, corrosion, or other foreign matter. Particular attention shall be given to the cavity of the body assembly, to insure that no grit, scale, or other foreign matter remains therein after final cleaning

3.6 Protective coating.-Protective coating shall be in accordance with the applicable drawing and the specifications referenced thereon.

3.7 Moisture content.

3.7.1 Tracer Composition - The moisture content of the tracer composition at th loading station at the time of loading the tracer compostion shall not exceed 0.10% when tested as specified in 4.3.3.1 & 4.4.1.

3.7.2 Igniter Composition.-The moisture content of the igniter composition at the loading station at the time of loading the igniter composition shall not exceed 0.14% when tested as specified in 4.3.3.1 & 4.4.1.

3.7.3 Initiator and relay ignition charge.-The moisture content of the black powder used in these charges at the loading station at the time of loading the initiator cup and the relay ignition charge shall not exceed 0.50% when tested as specified in 4.3.3.2 & 4.4.2.

3.7.4 Packaging boxes.-The moisture content of the packaging boxes prior to use shall not be more then 9.0% (see 4.3.3.3, 4.4.3 & 6.6).

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3.8 When the 20 packages of tracer assemblies are inserted into the barrier bag all excess air must be exhausted and the barrier bag is to be completely heat sealed (see 4.3.3.4, 4.4.4 & 6.7).

3.9 Tracers.-The tracers shall comply with all the requirements of Drawing D11751175 and with all the requirements specified in the applicable specifications. The tracers shall be free of dirt, chips, grease, rust, and other foreign matter. The cleaning method shall not be injurious to the metal nor shall the tracer be contaminated by the cleaning agents used.

3.10 Proving Ground test.-The tracer shall meet the following requirements, when tested as specified in 4.3.3.6 & 4.4.7.

a. Projectile shall not burst in gun bore or in flight within 400 yards of gun muzzle.
b. Function (produce a visible trace of light for a minimum period of 7.4 seconds).
c. Self-destruct.

4. QUALITY ASSURNACE PROVISIONS

4.1 Responsiblity for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification and drawing where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality Assurance terms and definitions. -Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.-Inspection shall be in accordance with MIL-A-2550, except as specified herein.

4.1.3 Inspection Criteria.

a. Successful passage of all inspections required by the drawings and specifications shall be considered as a condition precedent to performing the Proving Ground functional tests. (See 4.4.7.1).

b. Acceptance or rejection of a lot by the Government shall be contingent on successful passage of the Proving Ground functional tests.

4.2 First Article sample.

4.2.1 Initial production sample.-At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of assemblies and parts as follows:

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(Cont'd) 4.2.1

30 Tracer Assemblies, complete, including protective coating.
5 Tracer Assemblies, complete, including protective coating, minus tracer, igniter and initiator compositions.
5 Sets of component parts, uncoated and unassembled.

Thirty of the above complete tracer assemblies shall be forwarded to a designated proving ground for burn time testing and other characteristics listed in 3.10. The samples shall be manufactured using the same materials, equipment, processes and procedures, including packaging and packing as will be used in regular production and shall be obtained from the same source of supply. A first article sample or portion thereof as directed by the procuring agency shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacture processes, material used,drawing, specifications or source of supply, such as to significantly affect product uniformity as determined by the responsible technical agency.

4.2.1.1 Examination and test.-After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.-Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection.

4.3.1 Lot.

4.3.1.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Size of lot.-Unless otherwise specified by the procuring agency, a tracer assembly lot shall consist of an identifiable group of metal parts assemblie consisting of:

a. Metal parts produced by a homogeneous process
b. Initiator cup assemblies from not more than one lot
c. Relay ignition charge from not more than one lot
d. Tracer charge from not more than two lots consisting of materials from the same source.
e. Ignition composition from not more than one lot consisting of materials from the same source.

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4.3.2 Examination.-One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans contained in MIL-STD-105. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. An AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect shall be used. All nonconforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as cracks, burrs, and foreign matter. The roughness comparison specimens prescribed by ANSI-B46.1 shall be used as a basis for surface roughness determination.

4.3.2.1 Body, Drawing D11751178, covering a detail of Drawing F11751176.

	Defect	Method of Inspection
CRITICAL		
	None defined	
MAJOR		
101.	Pitch diameter of thread, min.	Gage
102.	Major diameter of thread, min.	Gage
103.	True position of outside diameter with major diameter of thread	Gage
104.	Diameter of charge cavity, min.	Gage
105.	Depth of charge cavity from disc and washer seat, min.	Gage
106.	True position of charge cavity with major diameter of thread	Gage
107.	Diameter of relay ignition charge hole	Gage
108.	True position of relay ignition charge hole with pitch diameter of thread	Gage
109.	Diameter of disc and washer cavity	Gage
110.	Depth to disc and washer seat	Gage
111.	Thickness crimp lip, min.	Gage
112.	Runout of rear taper at datum length with charge cavity	Gage
113.	Distance from disc and washer seat to rear shoulder	Gage
114.	Metal defective	Visual
115.	Protective coating incomplete or inadequate	Visual
MINOR		
201.	Outside diameter, min.	Gage
202.	Length from front face to forward shoulder, min.	Gage
203.	Width of thread undercut, max.	Gage
204.	Diameter of thread undercut	Gage
205.	Thickness from front face to bottom of charge cavity	Gage
206.	Length from front shoulder to wrench flats	Gage
207.	Distance across wrench flats	Gage
208.	Diameter at rear shoulder	Gage
209.	Raddi or chamfer, missing or incorrect	Visual (see 6.1)
210.	Burr	Visual
211.	Threads damaged or not full	Visual
212.	Foreign matter	Visual

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4.3.2.2 Disc, Closing, Drawing B11751180, covering a detail of Dwg. D11751175.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Diameter	Gage
102. Thickness, max.	Gage
103. Foreign matter	Visual
MINOR	
None defined	

4.3.2.3 Washer, Closing, Drawing B11751181, covering a detail of Dwg. D1175117

CRITICAL	
None defined	
MAJOR	
101. Outside diameter	Gage
102. Inside diameter	Gage
103. Total thickness, including parallelism	Gage
104. Thickness, min.	Gage
MINOR	
None defined	

4.3.2.4 Cup, Initiator, Dwg. C11751179, covering a detail of Dwg. C11751177.

CRITICAL	
None defined	
MAJOR	
101. Inside diameter, min.	Gage
102. Outside diameter, max.	Gage
103. Overall length, min.	Gage
104. Wall thickness at front end, max.	Gage

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4.3.2.4 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Radii or chamfer, missing or incorrect	Visual (See 6.1)
202. Burr	Visual
203. Foreign matter	Visual

4.3.2.5 Initiator Cup Assembly, Drawing C11751177, covering a detail of Drawing F11751176.

CRITICAL

None defined

MAJOR

101. Relay initiator charge missing or cracked	Visual
102. Relay initiator charge more than max. required depth from front face of initiator cup.	Gage

MINOR

None defined

4.3.2.6 Loading Assembly (Prior to inserting tracer charge and igniter composition). Drawing F11751176, covering a detail of Dwg D11751175.

CRITICAL

None defined

MAJOR

101. Initiator cup assembly missing or inverted	Visual

MINOR

None defined

4.3.2.7 Loading Assembly (after insertion of Tracer charge and igniter composition & prior to insertion of closing disc and closing washer), Drawing F11751176, covering a detail of Drawing D11751175.

CRITICAL

1. Relay initiator charge above front face of body	Gage

MAJOR

101. Relay initiator charge missing or cracked	Visual

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102.	Relay initiator charge more than max. required depth from front face	Gage

MINOR

201.	Igniter charge chipped or broken on any one step in excess of 1/3 the total area of the step, or total aggregate of chipped or broken area of all steps in excess of 1/3 of the total area of the largest step.	Visual

4.3.2.8 Tracer Assembly, Drawing D11751175

CRITICAL

1.	Tape missing or completely fails to cover relay ignition charge	Visual

MAJOR

		Gage
101.	Pitch diameter of thread, max.	Gage
102.	Outside diameter of shoulder, max.	Gage
103.	Overall length, max.	Gage

MINOR

201.	Disc missing, loose, punctured or torn	Visual/Manual
202.	Washer missing	Visual
203.	Crimp not full for 360 degrees	Visual
204.	Absence of adhesive	Visual
205.	Protective coating of tracer body damaged during loading	Visual

4.3.2.9 Fiberboard or Set-up box (prior to sealing)(see 5.1)

CRITICAL

None defined

MAJOR

None defined

MINOR

201.	Any Partitions (separators) missing	Visual
202.	Partitions (separators) damaged to the extent that tracer assemblies touch each other	Visual
203.	Tracer assembly missing	Visual

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4.3.2.10 Fiberboard or Set-up box (sealed) (see 5.1)

CRITICAL

None defined

MAJOR

101.	Sealing strip torn, badly wrinkled, incomplete or otherwise fails to seal box completely	Visual
102.	Box damaged exposing contents	Visual

MINOR

201.	Marking missing or unidentifiable	Visual (see 5.1)
202.	Contents loose	Manual

4.3.2.11 Barrier Bag (prior to sealing) (see 5.1)

CRITICAL

None defined

MAJOR

101.	No foreign matter or liquid in empty barrier bag	Visual
102.	Incorrect number of packages	Visual
103.	Bag, punctured, torn or cut	Visual

4.3.2.12 Barrier Bag (after exhausting excess air and sealing) (see 5.1)

CRITICAL

None defined

MAJOR

101.	Seal improper or incomplete	Visual
102.	Bag, punctured, torn or cut exposing contents	Visual

MINOR

201.	Marking missing or unindentifiable	Visual
202.	Contents loose	Manual

4.3.2.13 Packing box (prior to sealing)(See 5.1)

CRITICAL

None defined

MAJOR

101.	Sealed bag missing	Visual
102.	Warning Instruction sheet missing, (SMUFA Form 1519)	Visual

MINOR

None defined

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4.3.2.14 Packing Box (after sealing) (see 5.1)

CRITICAL

None defined

MAJOR

101.	Box broken, board loose or protruding nail.	Visual
102.	Strapping missing, improperly engaged, loose or broken	Visual

MINOR

201.	Marking missing or unidentifiable.	Visual
202.	Contents loose	Manual

4.3.3 Testing

4.3.3.1 Moisture and total volatile content of tracer composition and igniter composition.-The contractor shall provide adequate controls to assure the compliance of each lot with the applicable requirement. The contractor shall test a minimum of one randomly selected sample from each batch or from each 8 hours production, whichever is the lesser amount. Composite samples shall not be used for determining compliance. Failure to comply with the applicable requirement shall be cause for rejection of the parent quantity represented by the sample. Testing shall be performed as specified in 4.4.1 (see 3.7.1 & 3.7.2).

4.3.3.2 Moisture content of the initiator and relay ignition charges. The contractor shall provide adequate controls to assure compliance of each lot with the applicable requirement. The contractor shall test a minimum of one randomly selected sample from each batch or from each 8 hours production, whichever is the lesser amount. Composite samples shall not be used for determining compliance. Failure to comply with the applicable requirement shall be cause for rejection of the parent quantity represented by the sample. Testing shall be performed as specified in 4.4.2 (see 3.7.3).

4.3.3.3. Moisture content of packaging boxes.-The test for the moisture content of the packaging boxes shall be as specified in 4.4.3 (see 3.7.4 & 6.6).

4.3.3.4 Heat seal test for the barrier bag.-A heat seal test for the barrier bag shall be as specified in 4.4.4 (see 3.8 & 6.7).

4.3.3.5 Residual stresses-Cup, initiator-Sub-lots of initiator cups shall be tested for residual stresses as specified in 4.4.5 (see 3.3).

4.3.3.6 Proving Ground

4.3.3.6.1 Functioning.-(see 3.10).-This test shall be performed as specified in 4.4.7.1.

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4.3.3.6.1.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the applicable requirements specified, the sampling plan and procedure for this test shall be in accordance with MIL-STD-414, Section B-3, Code letter I and AQL 1.50 percent. In addition, 20 sample tracers shall randomly be selected from each of the first three lots, and fired for information only as specified in 4.4.7.1. If any tracer fails to ignite or fails to produce visible evidence of burning and self-destruct of the projectile, the lot shall be rejected.

4.3.3.6.1.2 After three consecutive lots have met the criteria of 4.3.3.6.1.1 the sampling plan and procedure for this test shall be in accordance with MIL-STD-414, Section B, Table B-3, Code letter F and AQL 1.50 percent. If any tracer fails to ignite or produce visible evidence of burning and self-destruct of the projectile the lot shall be rejected. (see 4.4.7.2)

4.3.3.6.1.3 Refiring.-If for any reason the proof director considers that the test conditions have detrimentally affected the test results, he shall declare the results invalid and perform the examinations and tests prescribed herein.

4.3.3.7 Inspection equipment.-LI-11751175 identifies the inspection equipment required to perform the examinations and tests prescribed herein.

4.4 Test methods and procedures.

4.4.1 Determination of moisture and total volatile content of tracer composition and igniter composition.-The mositure and total volatile content of the tracer composition and igniter composition shall be determined in accordance with MIL-STD-1234, Method 102.2 (see 3.7.1, 3.7.2, & 4.3.3.1).

4.4.2 Determination of moisture content of the initiator and relay ignition charges.-The moisture content of the black powder in these charges shall be determined in accordance with the procedure specified in MIL-P-223 (see 3.7.3 & 4.3.3.2).

4.4.3 Determination of moisture content of the packaging boxes.-A 2 inch by 2 inch sample square shall be cut from a packaging box. This sample shall be weighed and the weight recorded. Then the sample shall be placed in an oven for 4 hours at a temperature of 150 degrees Fahrenheit (150°F). After this the sample shall be reweighed to accept or reject on the results (see 3.7.4, 4.3.3.3 & 6.6).

4.4.4 Failure of the heat seal test of the sealed barrier bag is a major defect. Sampling, acceptance criteria and testing procedures for this test shall be as required in specification MIL-P-116 (see 3.8, 4.3.3.4 & 6.7).

4.4.5 Residual stresses - Cup, Initiator.-The purpose of this mercurous nitrate test is to detect residual stresses in brass components which could result in cracks or splits during extended ammunition storage. Lot size to be 100,000 pieces maximum. Divide the lot into Sub-lots of 10,000 pieces equally and subject a sample of 25 pieces of each sub-lot to this test. The sub-lot shall be rejected if one or more initiator cups of the sample shows a split.

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4.4.5.1 Mandatory safety requirememts.

4.4.5.1.1 Food shall not be eaten or stored in the laboratory or space in which these tests are conducted.

4.4.5.1.2 Acid resistant apron and gloves or the equivalent shall be worn by each test technician.

4.4.5.1.3 Face shields shall be worn at all times during the pouring or mixing of acids and water. Safety glasses shall be worn during other phases of this test.

4.4.5.1.4 Asbestos or heat insulating gloves shall be worn during the heat volatilization phase of the test to facilitate handling.

4.4.5.1.5 During the entire period of volatilization the oven door shall be closed.

4.4.5.1.6 Extreme care shall be exercised in the mixing of acid with water and shall only be accomplished by the pouring of the acid into the water.

4.4.5.1.7 This test shall be conducted under canopy or hood having a forced draft ventilation system to remove the noxious fumes. In the disposition of tested components local regulations shall be observed.

4.4.5.2 Preparation for test. The initiator cups are submerged in an aqueous solution of nitric acid (40% by volume of 1.42 specific gravity nitric acid in distilled water) for thirty seconds and are then rinsed thoroughly in cold running water.

4.4.5.3 Method of conducting test.

4.4.5.3.1 The aqueous mercurous nitrate solution used in this test shall consist of 10 grams of mercurous nitrate and 10 milliliters of nitric acid (spec. gr. 1.42) per liter of solution.

4.4.5.3.1.1 The aqueous mercurous nitrate solution container shall be equippe with a suitable grid or screen in the bottom to permit full access of the solution to the head of the initiator cup. The level of the solution in the container shall be such that the initiator cup is completely submerged when standing upright on the grid or screen.

4.4.5.3.2 The initiator cups shall then be immersed upright in the mercurous nitrate solution for 15 minutes. They are then rinsed thoroughly in cold runnin water and drained.

4.4.5.3.3 After completion of the procedure of 4.4.5.3.2, the initiator cups are then placed in an oven or other suitable heat source to volatilize the mercury.

4.4.5.3.4 When the mercury has been completely volatilized, the initiator cups are examined under a magnification of 10 to 15 power for cracks or splits.

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4.4.5.4 Definitions

4.4.5.4.1 A split is a separation of metal entirely through the wall or head of the initiator cup.

4.4.5.4.2 A crack is a surface condition in which the separation of the metal does not go entirely through the wall or head of the initiator cup.

4.4.5.5 Recording of data.

4.4.5.5.1 The results of the test shall be recorded (See 4.4.5.6).

4.4.5.5.2 The test record shall be reproduced and copies distributed as directed by the responsible agency for the acceptance of the item (See 6.13).

4.4.5.6 Suggested Format

MERCUROUS NITRATE TEST - ACCEPTANCE REPORT

End Item: - Tracer, Projectile Assembly, M3A1
Part Name: - Cup, Initiator
Part No.: - 11751179
Contractor: -
Contract No.: -
Lot: -
Sub-Lot No.: -
Sub-Lot Size: -

Total No. Tested_____

SPLITS Results_____
CRACKS Results_____

Remarks:

_____
Foreman

_____
Test Technician

_____
Chf, Ballistics

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4.4.6 Protective coating.- The procedures for testing the cleaning of component and the application of protective coating shall be in accordance with the criteria for testing, as specified in the applicable specification. The salt spray test shall be performed utilizing the procedures and equipment prescribed by ASTM Method B117.

4.4.7 Proving Ground.

4.4.7.1 Ballistic tracer functioning (Acceptance test).- The sample tracers shall be assembled to HE-T, SD loaded Mk 2 40MM cartridges. Unless otherwise instructed by the technical agency, immediately prior to assembling the sample tracers to the projectiles, the tape protecting the relay ignition charge shall be removed. The assembled rounds shall be fired single action at ambient temperatur from the appropriate weapon and the tracer burning times measured and recorded. A timing device capable of measuring time to the nearest .01 second shall be used to measure burning time. Suitable aides, such as binoculars, should be used to insu visible contact with tracer (See 6.12 & 6.13).

4.4.7.2 Ballistic tracer functioning (information only).- The sample tracers shall be assembled to HE-T, SD loaded Mk 2 40MM cartridges in the manner as specifie in 4.4.7.1 and tested as specified in 4.4.7.1. One-Half of the assembled test round shall be temperature conditioned at 125 degrees Fahrenheit (125°F) for a minimum of 18 hours and those remaining shall be temperature conditioned at minus 40 degrees Fahrenheit (-40°F) for a minimum of 18 hours. The assembled rounds shall then be fired and tracer burning time recorded for informational purposes only (See 6.12 & 6.

4.4.8 Government use of contractor's inspection equipment.- The contractor shall make available to the Government all inspection equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment and verification of its accuracy, shall be supplied by the contractor for the performance of evaluation or test by the Government.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with Picatinny Arsenal Provisional Packing Instruction No. 92-74, dated 22 October 1974 (sheets 1,2 and 3 of 3 sheets). These provisional packing instructions shall be utilized as part of the basic procurement documents until advised of a substitution by the technical agency.

6. NOTES

6.1 Visual examination qualification.- When compliance with the applicable requirement is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

6.2 Visual standards.- Visual standards to be established at contractor's plant in accordance with 3.4.3.

6.3 Ordering data.- Procurement documents should specify the title, number, and date of this specification.

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6.4 Moisture.- Igniter and tracer compositions.-The pellets formed thereof are hygroscopic and will deteriorate rapidly when exposed to moisture. Small amounts of moisture present in or absorbed by the composition or pellets, prior to or after loading into tracers will eventually result in unsatisfactory functioning. It is highly desirable that the ingredients, compositions, pellets, and loaded tracers be carefully safeguarded from exposure to moisture and humid atmospheres. For assurance of a satisfactory loaded tracer, it is recommended that the blending and pelleting buildings be air conditioned in a manner which will hold the humidity of the atmosphere as low as practicable.

6.5 Moisture.-Initiator charge and relay ignition charge (Black Powder). Small amounts of moisture present in or absorbed by the black powder prior to or after loading the initiator cup and tracer assembly will result in unsatisfactory tracer functioning. It is highly recommended that the black powder, loaded initiator cups and tracer assemblies be carefully safeguarded from exposure to moisture and humid atmosphere.

6.6 To assure compliance of 3.7.4, 4.3.3.3 & 4.4.3 it is recommended that the packaging boxes be stored in a cool dry storage area prior to use.

6.7 Heat Seal Test of barrier bag.- Verification of the heat seal of the barrier bag shall be as required in specification MIL-P-116.

6.8 Prior to fabrication of an initial production sample, the contractor upon approval of the contracting officer may make and submit for evaluation up to 5 process control lots. These process control lots shall consist of not less than 10 nor more than 30 tracer assemblies. The contractor shall furnish all information pertaining to the manufacture of the lots to the contracting officer and the technical agency. All costs for testing these process control lots will be borne by the Government.

6.9 Alternative testing procedures.- Alternative testing procedures may be used by the contractor when such testing and procedures provide as a minimum the quality assurance provisions specified herein. Prior to applying alternative procedures, the contractor shall describe them in a written proposal and shall demonstrate for approval of the responsible technical agency that their effectiveness is equal to or better than the quality provisions specified herein.

6.10 When a product is being inspected in accordance with MIL-STD-1235 and a defective characteristic is found, the product produced prior to the discovered defect will be screened back 2000 tracer assemblies and all defects culled out.

6.11 Shipping.-It is not intended to ship these compositions or pellets. It is intended that the compositions and pellets thereof be prepared at the plant where the loading of the tracers will be accomplished.

6.12 Combining of proving ground tests.- When the contractor for the tracer is also the contractor for the complete round or one of the loaded components thereof, the proving ground tests may be combined in order to save expense, upon agreement between the procuring activity and the contractor.

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6.13 Submission of mercurous nitrate, firing and test data.- In addition to the normal distribution of records for each lot of tracers, when the item is procured by the Department of Defense, copies should be sent to:

Commander
Frankford Arsenal
ATTN: SARFA-QAA-Q (1 copy)
SARFA-MDP-Y (1 copy)
Philadelphia, Pa. 19137

Custodian:

Army - AR

Preparing Activity

Army - AR

Project No. 1315-A261

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b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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